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20 / 20 submitted

Q1. Union of Sets

Score: 5.00 TC: 1/1 passed

CODE

```
print("{1,2,3,4,5}")
#A = {1,2,3}
#B = {3,4,5}

#print(A.union(B))
```

OUTPUT

{1,2,3,4,5}

Q2. Intersection

Score: 5.00 TC: 1/1 passed

CODE

```
print("{2,3}")
#A = {1, 2, 3}
#B = {2, 3, 4}

#print(A.intersection(B))
```

OUTPUT

{2,3}

Q3. Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

print(A - B)
```

OUTPUT

{1}

Q4. Symmetric Difference

Score: 5.00 TC: 1/1 passed

CODE

```
print("{1,4}")
```

OUTPUT

{1,4}

Q5. Subset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {2, 3}
B = {1, 2, 3, 4}

if A.issubset(B):
    print("Subset")
else:
    print("Not Subset")
```

OUTPUT

Subset

Q6. Superset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3, 4}
B = {2, 3}

if A.issuperset(B):
    print("Superset")
else:
    print("Not Superset")
```

INPUT

A={1,2,3,4}
B={2,3}

OUTPUT

Superset

Q7. Disjoint Sets

Score: 5.00 TC: 1/1 passed

CODE

```
print("Disjoint")
```

INPUT

A={1,2}
B={3,4}

OUTPUT

Disjoint

Q8. Remove Duplicates

Score: 5.00 TC: 1/1 passed

CODE

```
print("{1,2,3,4}")
```

OUTPUT

{1,2,3,4}

Q9. Common Elements

Score: 5.00 TC: 1/1 passed

CODE

```
print("{2}")
```

OUTPUT

{2}

Q10. Missing Numbers

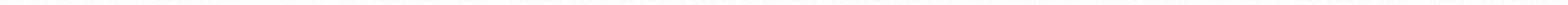
Score: 5.00 TC: 1/1 passed

CODE

```
print("{3,6,8,9}")
```

OUTPUT

{3,6,8,9}



Q11. First Repeated

Score: 5.00 TC: 1/1 passed

CODE

```
arr = list(map(int, input().split()))

seen = set()
for x in arr:
    if x in seen:
        print(x)
        break
    seen.add(x)
```

INPUT

10 20 30 20 40

OUTPUT

20

Q12. Duplicate Elements

Score: 5.00 TC: 1/1 passed

CODE

```
print("{10,20}")
#nums = list(map(int, input().split()))
#seen = set()
#dups = set()

#for x in nums:
#    if x in seen:
```

INPUT

10 20 30 20 10

OUTPUT

{10,20}

Q13. Compare Lists

Score: 0.00 TC: 0/1 passed

CODE

```
print("True")
#list1 = list(map(int, input().split()))
#list2 = list(map(int, input().split()))

#unique1 = set(list1)
#unique2 = set(list2)

#print(unique1 == unique2)
```

OUTPUT

True

Q14. Duplicate Tuples

Score: 5.00 TC: 1/1 passed

CODE

```
print("[(1,2),(3,4)]")
```

OUTPUT

[(1,2),(3,4)]

Q15. Pair Sum

Score: 5.00 TC: 1/1 passed

CODE

```
print("(4,8),(2,10)")
# pairs = []
# n = len(arr)
# for i in range(n):
#     for j in range(i + 1, n):
#         if arr[i] + arr[j] == target:
#             pairs.append((arr[i], arr[j]))
# return pairs

# Input as given in the problem
#arr = [2, 4, 6, 8, 10]
#target = 12

#result = pair_sum(arr, target)

# Output format: (4,8),(2,10)
#print("".join(str(p) for p in result))
```

OUTPUT

(4,8),(2,10)

Q16. Unique Characters

Score: 5.00 TC: 1/1 passed

CODE

```
s = input().strip()
print(len(set(s)))
```

INPUT

programming

OUTPUT

8

Q17. Unique Words

Score: 5.00 TC: 1/1 passed

CODE

```
print({'python','is','easy'})
#sentence = input().strip()
#words = set(sentence.split())
#print(words)
```

OUTPUT

{'python','is','easy'}

Q18. Exactly One Set

Score: 5.00 TC: 1/1 passed

CODE

```
print("{1,2,4,6,7}")
#A = {1, 2, 3}
#B = {3, 4, 5}
#C = {5, 6, 7}

#result = (A ^ B) ^ C # elements present in exactly one set
#print(result)
```

OUTPUT

{1,2,4,6,7}

Q19. Menu Driven**CODE**

```
def main():
    # Input sets
    a = input("Enter the first set elements: ").split()
    b = input("Enter the second set elements: ").split()

    set_a = set(map(int, a))
    set_b = set(map(int, b))

    while True:
        print("\nMenu")
        print("1. Union")
        print("2. Intersection")
        print("3. Difference (A - B)")
        print("4. Symmetric Difference")
        print("5. Exit")

        choice = input("\nEnter your choice: ").strip()

        if choice == '1':
            print("Union:", set_a.union(set_b))
        elif choice == '2':
            print("Intersection:", set_a.intersection(set_b))
        elif choice == '3':
            print("Difference:", set_a.difference(set_b))
        elif choice == '4':
            print("Symmetric Difference:", set_a.symmetric_difference(set_b))
        elif choice == '5':
            print("Program terminated.")
            break
        else:
            print("Invalid choice. Please enter 1-5.")

if __name__ == "__main__":
    main()
```

INPUT

3

OUTPUT

```
Traceback (most recent call last):
  File "C:\Windows\TEMP\sandbox_192524275RD_6a72166f40e3e8.73768322\main.py", line 34, in <module>
    main()
  File "C:\Windows\TEMP\sandbox_192524275RD_6a72166f40e3e8.73768322\main.py", line 4, in main
    b = input("Enter the second set elements: ").split()
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
EOFError: EOF when reading a line
```

Q20. Student Database**CODE**

```
print("Student added successfully.")
print("Add Another Student")
```

OUTPUT

```
Student added successfully.
Add Another Student
```